THE ALMOST-NORMAL DISTRIBUTION

1. Introduction

A 2014 work [2] of Shaw, Luu, and Brickman mentions a specific construction of the so-called 'almost-normal' distribution, having been defined for the purposes of an exam question. In this note, we recall its definition, and review some of its notable properties.

The distribution itself is defined implicitly by the following algorithmic procedure:

Algorithm 1 Generating Almost-Normal Random Variables

(1) Sample $U \sim \mathsf{Unif}\left([-1,1]\right)$.

(2) Set $X = \operatorname{sign}(U) \cdot \sqrt{2 \cdot \log\left(\frac{1}{1-U^2}\right)}$.

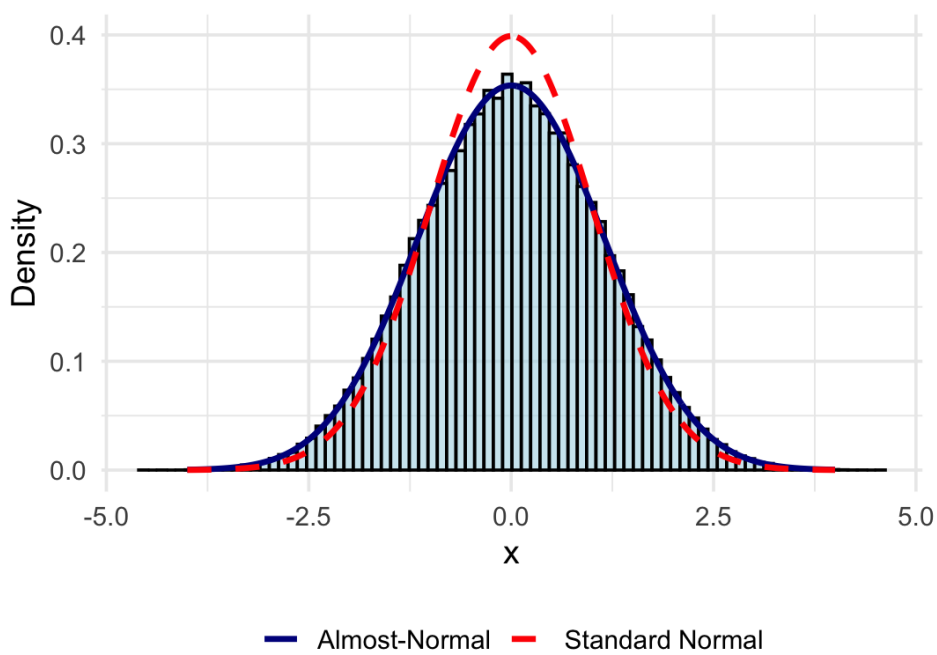
2. Basic Properties

Given this, define ν to be the law of X. With a little bit of work, one can establish that ν may be written as

$$\nu\left(\mathrm{d}x\right) = \frac{1}{4} \cdot \frac{|x|}{\left(1 - \exp\left(-\frac{|x|^2}{2}\right)\right)^{1/2}} \cdot \exp\left(-\frac{|x|^2}{2}\right) \mathrm{d}x,$$

where the density at 0 is seen to be finite by e.g. l'Hôpital's rule.

Figure 2.1. Plot of Density for Standard Normal and Almost-Normal



Adopting the language of statistical physics, the 'potential' associated to this measure is given (up to a constant) by

$$V\left(x\right) = \frac{3}{8}\left|x\right|^2 + \frac{1}{2} \cdot \log \operatorname{sinhc}\left(\frac{|x|^2}{4}\right),$$

with $\operatorname{sinhc}(z) = z^{-1} \cdot \sinh z$.

3. Some Derived Properties

One may check that this V is strongly-convex, with strong convexity constant equal to $\frac{3}{4}$. That one may take $m \geq \frac{3}{4}$ follows from the convexity of $x \mapsto \log \operatorname{sinhc}\left(\frac{|x|^2}{4}\right)$, and that taking $m < \frac{3}{4}$ is sub-optimal follows from examining the behaviour of V around $x = 0$. In particular, it follows that ν is strongly log-concave, and hence satisfies various appealing functional inequalities.

Using general results due to Bobkov and Houdré [1], the isoperimetric profile of ν can be computed explicitly as

$$I_\nu(p) = p \cdot (1-p) \cdot \left(\frac{2}{(1-2\cdot p)^2} \cdot \log\left(\frac{1}{4 \cdot p \cdot (1-p)} \right) \right)^{\frac{1}{2}}, \qquad p \in [0,1].$$

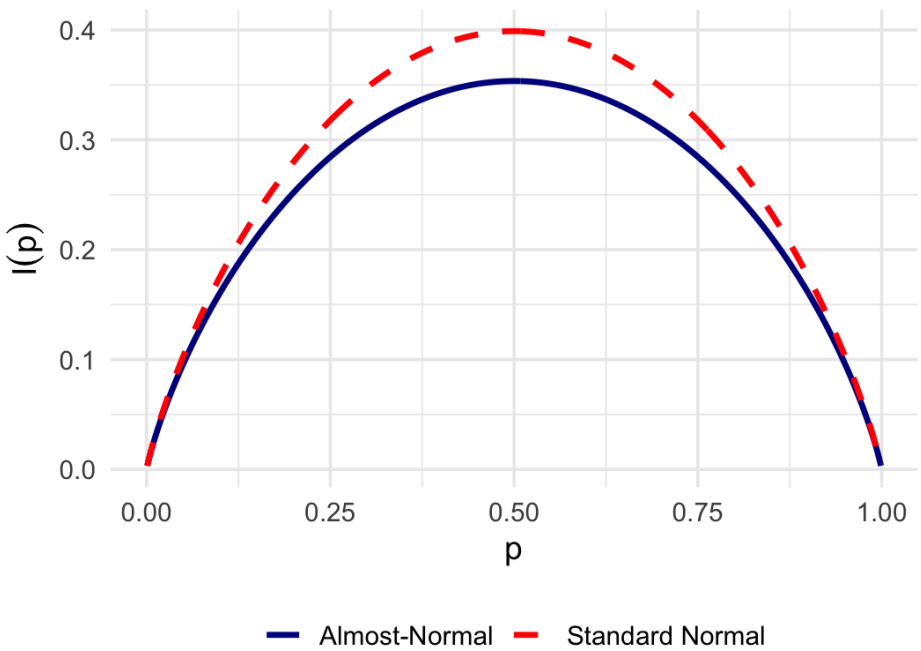
Observe that as $p \to 0^+$, one has the asymptotic behaviour

$$I_\nu(p) \sim p \cdot \left(2 \cdot \log\left(\frac{1}{4 \cdot p} \right) \right)^{\frac{1}{2}},$$

which is to be compared with the isoperimetric profile of the standard normal,

$$I_\gamma(p) \sim p \cdot \left(2 \cdot \log\left(\frac{1}{p} \right) \right)^{\frac{1}{2}}.$$

Figure 3.1. Plot of Isoperimetric Profile for Standard Normal and Almost-Normal

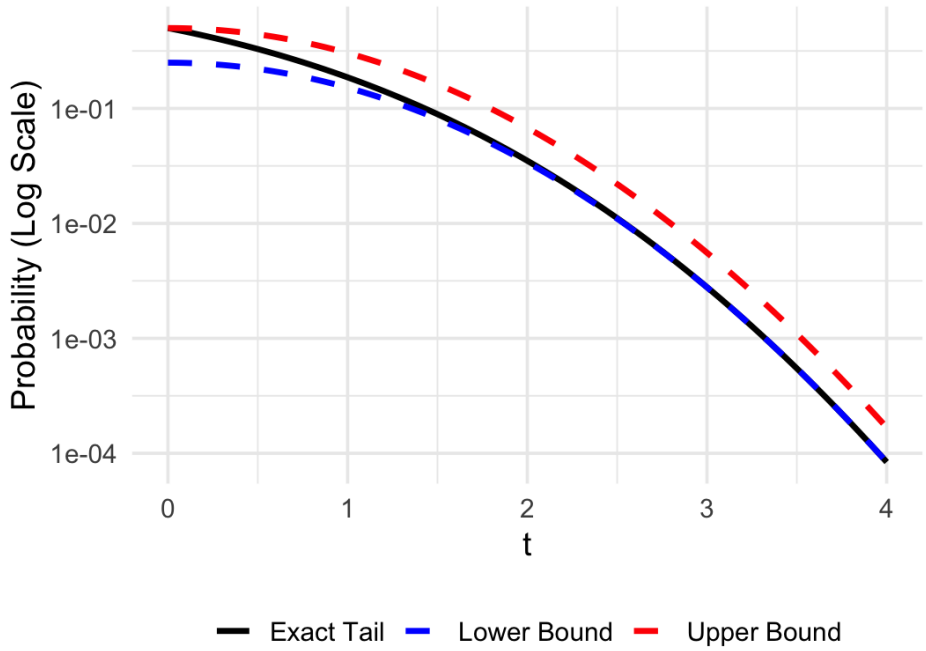


Additionally, for $t \geq 0$, one can compute that

$$\begin{aligned} \nu((t,\infty)) &= \frac{1}{2} \cdot \left(1 - \left(1 - \exp\left(-\frac{t^2}{2} \right) \right)^{\frac{1}{2}} \right) \\ &\in \left[\frac{1}{4}, \frac{1}{2} \right] \cdot \exp\left(-\frac{t^2}{2} \right), \end{aligned}$$

where the lower limit is asymptotically sharp for large t, and the upper limit is asymptotically sharp for small t.

FIGURE 3.2. Plot of Tail Probability and Bounds for the Almost-Normal



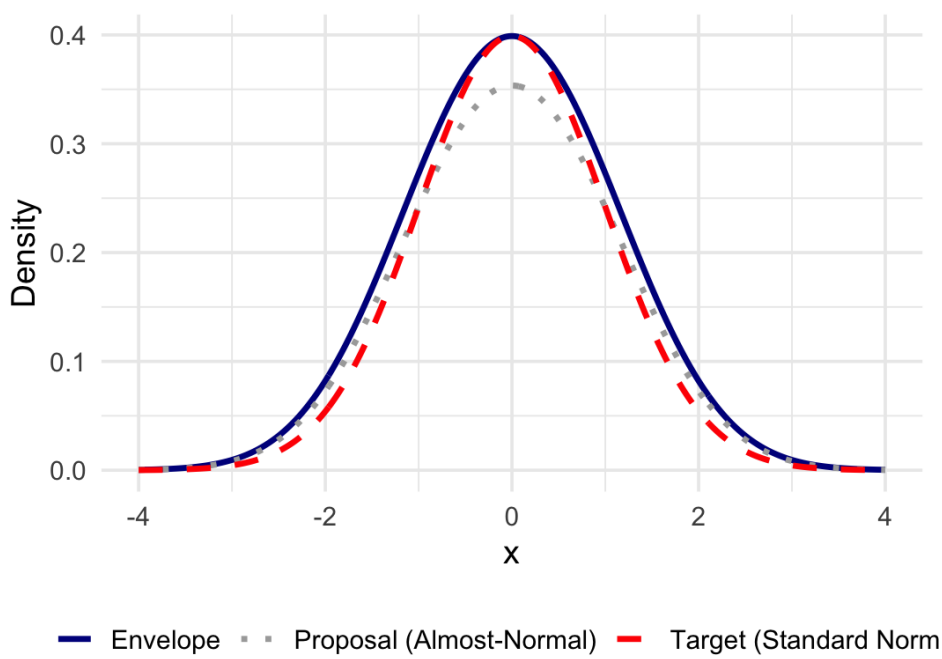
4. Application to Rejection Sampling

From the point of view of rejection sampling, it bears mentioning that there holds the bound

$$\left\| \frac{\mathrm{d}\gamma}{\mathrm{d}\nu} \right\|_\infty = \left(\frac{4}{\pi} \right)^{\frac{1}{2}} \leq 1.13,$$

suggesting a high level of efficiency as a proposal distribution for the standard normal.

FIGURE 4.1. Plot of Almost-Normal as Rejection Sampling Envelope for Standard Normal



One might consider using a rescaled version of the almost-normal distribution as a proposal in the same context, i.e. proposing with $\sigma \cdot X$ for some other $\sigma > 0$. Note that for $\sigma < 1$, the tails are too light to obtain a valid envelope, and so we focus on $\sigma \geq 1$. The corresponding density is then given by

$$\nu_\sigma\left(\mathrm{d}x\right) = \frac{1}{4\sigma^2} \cdot \frac{|x|}{\left(1 - \exp\left(-\frac{|x|^2}{2\sigma^2}\right)\right)^{1/2}} \cdot \exp\left(-\frac{|x|^2}{2\sigma^2}\right) \mathrm{d}x.$$

Examining the behaviour around $x = 0$, one sees that

$$\left\| \frac{\mathrm{d}\gamma}{\mathrm{d}\nu_\sigma} \right\|_\infty \geq \frac{\mathrm{d}\gamma}{\mathrm{d}\nu_\sigma}\left(0\right) = \left(\frac{4}{\pi}\right)^{\frac{1}{2}} \sigma,$$

and so taking $\sigma > 1$ cannot lead to a tighter envelope. As such, in some local sense, the almost-normal proposal cannot be improved.

5. Conclusion

In the end, the claim that this distribution is 'almost normal' is indeed borne out in various comparisons.

References

[1] Serguei Germanovich Bobkov and Christian Houdré. *Some connections between isoperimetric and Sobolev-type inequalities*, volume 129. American Mathematical Soc., 1997.

[2] William T Shaw, Thomas Luu, and Nick Brickman. Quantile Mechanics II: changes of variables in Monte Carlo methods and GPU-optimised normal quantiles. *European Journal of Applied Mathematics*, 25(2):177–212, 2014.